AMRITA VISHWA VIDYAPEETHAM
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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I007*

Team Name: *Ryvex*

Domain: *Intelligent Systems and Digital Futures*

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Presentation Outline

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Problem Statement

Imagine that two industries are near to each other. One industry produces heat waste while other consumes gallons of water to cool down. Because both industries are not used properly they are often seen as pollution since one uses gallons of water and the other causes pollution. This is where our idea comes in. **Our digital twin is designed to turn heat vapor from industries into water that can be useful for cooling down data centers.**

Problem Understanding

- **Who experiences it?**
  - Data centre operators, those running AI/GPU servers and shared colocation sites
  - Industrial sites such as textile mills, paper factories, chemical plants and foundries that generate waste heat
- **What causes it?**
  - **Heavy water use:** Data centers uses gallons and gallons of water to cool down
  - **Waste-heat pollution:** factories releases waste heat into the air in the form of plume instead of reusing it
  - **Lack of market:** no common method to price or trade excess heat and reclaimed water
- **How does it affect stakeholders?**
  - Data centres: Very high electricity costs and heavy use of city water a problem in busy cities like Chennai and Bengaluru
  - Industry: reusable energy is wasted which is capable of generating income
  - Society: strain on water supplies plus emissions that could have been prevented

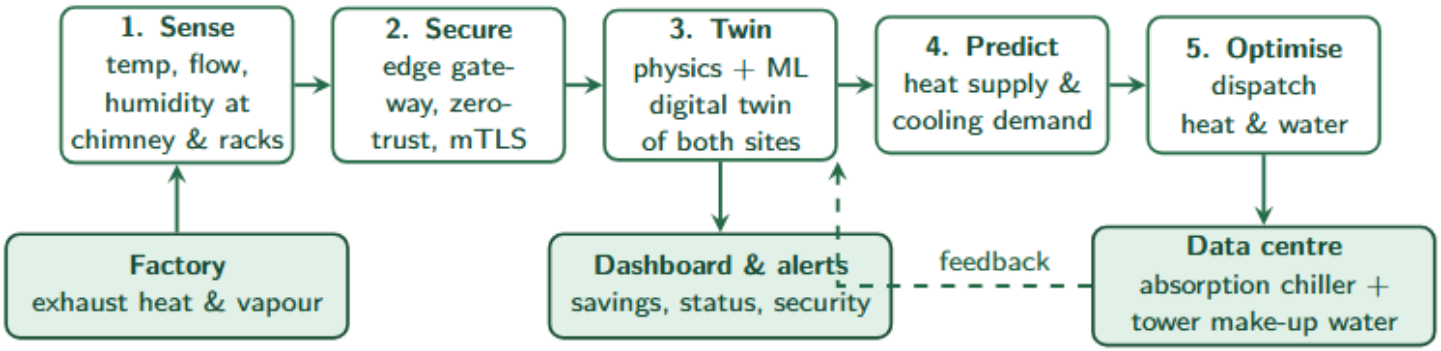
Research Gap: Existing Solutions vs RYVEX

Area	Existing	RYVEX
1. Data-centre cooling	Conventional and advanced cooling systems focus mainly on maintaining server temperatures and improving cooling efficiency.	Proposes AI forecasting and digital-twin simulation to recommend efficient cooling strategies based on changing workloads and weather conditions.
2. Industrial waste-heat recovery	Waste heat is recovered for specific applications, such as heating or thermally driven cooling.	Aims to coordinate available industrial waste heat with data-centre cooling demand.
3. Water recovery	Water-recovery approaches generally focus on specific sources and processes.	Explores recovering moisture from suitable exhaust streams for potential reuse after treatment and quality checks.
4. Resource optimisation	Individual systems may optimise cooling energy, heat recovery, or water consumption separately.	Proposes evaluating energy, water, heat utilisation, and equipment reliability together.
5. Predictive decision-making	Monitoring and optimisation may focus on individual facilities or systems.	Combines forecasts of industrial heat availability, data-centre workloads, cooling demand, and weather to recommend operating strategies.
6. Performance validation	Cooling and heat-recovery technologies have established evaluation methods.	Needs comparative simulation and practical testing to determine whether its integrated approach reduces energy and water consumption while maintaining reliability.

Our Proposed Solution

- **Core idea:** Ryvex, an AI-driven digital-twin that can convert industrial waste heat to water and can be used for cooling data-centres.
- **How it works: Sense → Secure → Twin → Predict → Optimise.**
  - Sensors monitor temperature, steam and cooling needs; a zero-trust gateway keeps factory systems separate.
  - An AI-Forecasting and resource optimization twin predicts what is needed and what is available; the optimiser directs heat to absorption chillers. Reused water, to cooling towers.
- **Value delivered (targets):**
  - Data centres: 20–30% less fresh water, 10–15% less cooling power
  - Factories: Earns income from heat and vapour that were previously vented.
  - Cities: They experience water stress, lower grid demand and lower emissions.

Technical Approach



Sensors gather information AI figures out when heating or cooling is needed and the digital twin makes sure resources are used properly. A dashboard shows the results and safety controls make sure everything works safely.

Tech Stack

Frontend: Streamlit

Backend: Python, FastAPI

Database: PostgreSQL/TimescaleDB

AI/ML, APIs, or Hardware: Scikit-learn, Pandas, NumPy, SciPy/PuLP, simulated IoT sensors, MQTT, and digital-twin simulation

Innovation & Uniqueness

- **What is new:**
  - A single platform that exchanges both heat and water between industry and data centres
  - AI-based forecasting to predict future demands
  - Security by design: a zero-trust connection between factory OT and data centre IT
- **Why it fits the problem:** Supply and demand often change at times and rarely match up. That's why forecasting and real-time dispatch are key to closing the timing gap. The digital twin allows every decision to be tested first.
- **Key differentiator:** To our knowledge this is the system that brings together forecasting, dynamic heat and water dispatch and OT–IT security, in one low-cost open-source retrofit-friendly solution.

Feasibility & Scalability

- **Feasibility:** This plan uses ready-made sensors, small local computers and open-source software.
  - Phase 1: Train the twin on simulated data.
  - Phase 2: Validate on a lab heat-exchanger rig.
  - Phase 3: Pilot at a partner site near a data centre.
- **Scalability:** The system uses nodes; Scalability means we can add more plants, more data centres or even a whole industrial park and then copy the model to those new places.
- **Risks and mitigations:**
  - **Heat loss over distance:** The risk is that heat can drop when it travels far. The mitigation is to choose sites that're close together or co-located.
  - **Water Quality:** The risk is that water may become dirty. The mitigation is to treat the water and also to use quality sensors before reuse.
  - **Cyber risk:** The risk is that the network could be attacked. The mitigation is to use zero-trust access and to split the network into segments.
  - **Partner willingness:** The risk is that partners may not want to join. The mitigation is to share the revenue, from waste heat so they have an incentive.

Cost for Product

RYVEX: Total Product Cost Estimate

Recommended initial prototype budget: INR 40,000
For a working student prototype with a dashboard, AI forecasting, digital-twin simulation, and basic sensor integration.

Component	Cost
AI model and digital-twin simulation	INR 8,000
Sensors and data acquisition	INR 10,000
Microcontroller and connectivity	INR 5,000
Dashboard and software setup	INR 5,000
Assembly and testing	INR 5,000
Contingency and miscellaneous	INR 7,000
Total estimated cost	**INR 40,000**

Estimated cost at different stages

- **Software-only demonstration:** INR 5,000–15,000
- **Working prototype with sensors:** INR 25,000–60,000
- **Industrial pilot:** INR 5 lakh–15 lakh or more, depending on integration and equipment requirements
- **Full industrial deployment:** requires a site-specific engineering quotation; major cooling and heat-recovery equipment can substantially increase the cost

Use Cases

- Use case 1: **Plant operator** (textile or paper mill). Exhaust heat that was previously vented is now sent to a data centre. There it powers an absorption chiller. Result: the neighbouring data centre uses energy for cooling and the original site earns new revenue, from the heat sale. **Outcome:** new revenue and lower cooling demand next door.
- Use case 2: **A Data centre operator** runs AI workloads. Ryvex predicts GPU load spikes and pre-adjusts cooling using factory heat and water. **Outcome:** less grid power and tap water.
- Use case 3: **Industrial park or city authority.** A dashboard keeps track of the shared heat and water exchange. Checks the savings and security. The dashboard is important because it shows how much is saved and if everything is secure. It helps people understand the benefits and the safety of the system. The dashboard is a tool, for monitoring and verifying the results. It makes sure that the savings are real and that the security is strong. The dashboard helps to keep everything easy to follow. It is a part of the system that provides important information. **Outcome:** auditable sustainability reporting.

Impacts and Benefits

- **Benefit to users:** Data centres reduce cooling costs and depend less on city water; factories make money from waste heat and steam.
- **Social, economic and environmental impact:**
  - Social: makes it easier to reduce competition for water in cities that have water problems
  - Economic: reduces operating costs creates a new way to make money supports greener industrial parks
  - Environmental: uses less fresh water uses less power from the grid produces fewer emissions
- **Expected measurable outcomes (targets):**
  - 20–30% reduction in freshwater use
  - 10–15% reduction in cooling power
  - Forecast error below 10%
  - Zero unauthorised OT access in testing

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 6	Target 6.4	Clean Water and Sanitation	Recovered water cuts freshwater use in cooling
SDG 7	Target 7.3	Affordable and Clean Energy	Waste heat replaces electric chiller load
SDG 9	Target 9.4	Industry, Innovation and Infrastructure	Retrofits industry and data centres for resource efficiency
SDG 12	Target 12.2	Responsible Consumption and Production	Reuses heat and water instead of wasting them

Individual Member Contribution

Member	Contribution or responsibility
Rhaghavi S N	Team lead; storyline, research and dashboard
Charu Nethra C	Security design and data pipeline
Sai Durga C	Forecasting models and optimiser
Bhavya V	Digital-twin modelling

Yukti ID

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